



Blockchain Technologies and Corporate Governance Reformation: An Agency Theory Perspective

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Keywords:	blockchains, permissioned blockchains, data analytics, data analysis, Smart Cites, Governance, Planning and Policy
Abstract:	Technology has been indispensable in the present age for survival and competitive advantages of organizations. Among the technologies shaping corporate governance recently is blockchain technology. Five research objectives were examined by this research, which include the bibliometric and thematic analysis of existing relevant literature, how blockchain has been included in corporate governance, how this mitigates agency problems, and the associated impacts and risks. To achieve the objectives, secondary data technique was employed. Data was collected from the SCOPUS academic database using the PRISMA model. 75 articles were found relevant, which formed the dataset for this research. Software such as Microsoft Excel, VOS viewer, Harzing's publish or perish, and LaTeX coding using Overleaf was used for data analysis. The findings show that there are limited existing studies. Themes that have emerged include smart contracts, DAOs, and corporate voting, among others. Also, an in-depth discussion of the agency perspective, impact, and risks of blockchain in the corporate world was presented. This study is relevant to various stakeholders on blockchain inclusion in corporate governance. Moreover, there is a need for increased legal support and corporate regulations on the usage of disruptive technologies in organizations. Future studies should conduct empirical studies in this area.

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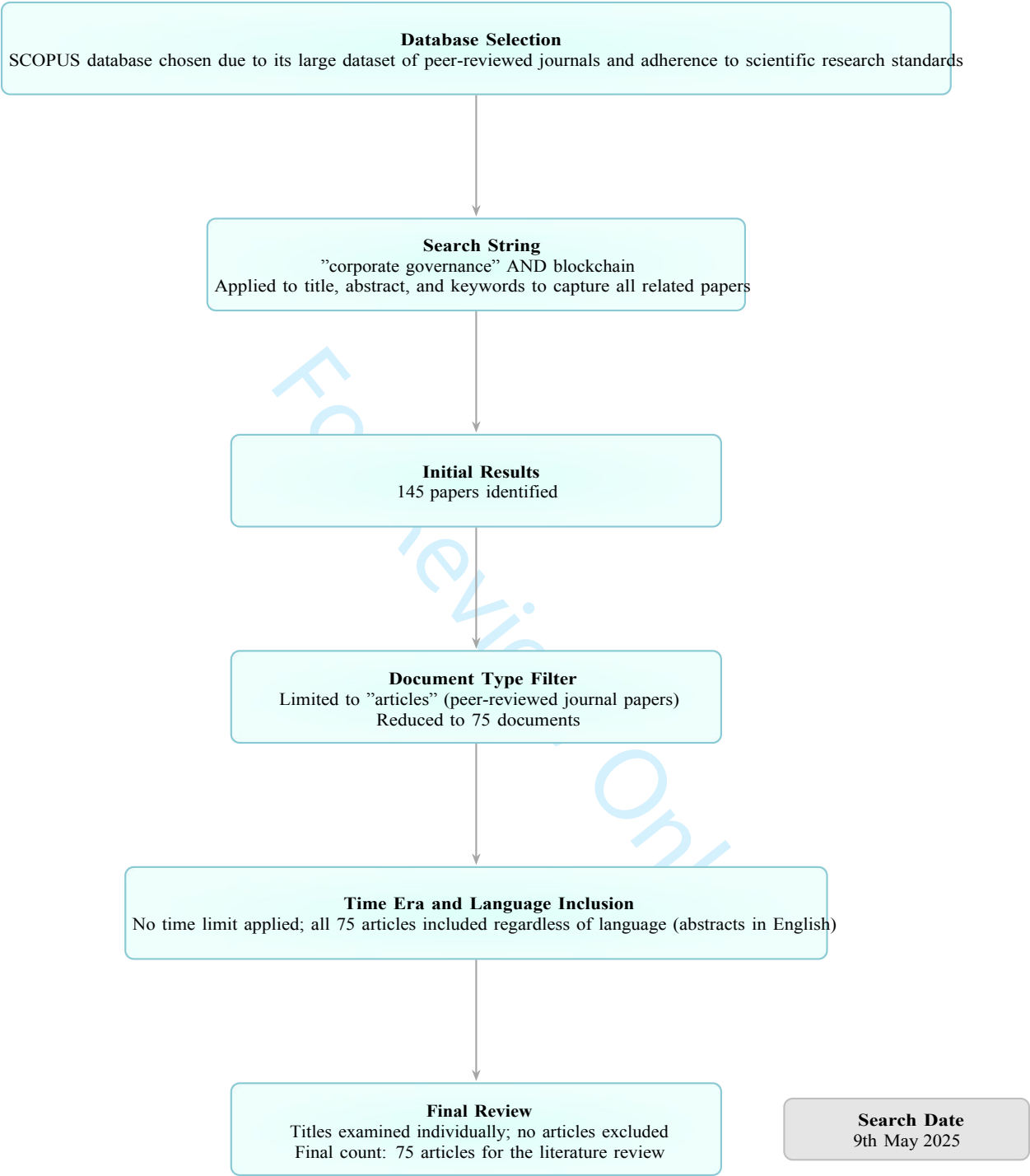


Figure 1: PRISMA Flow Diagram for Methodology

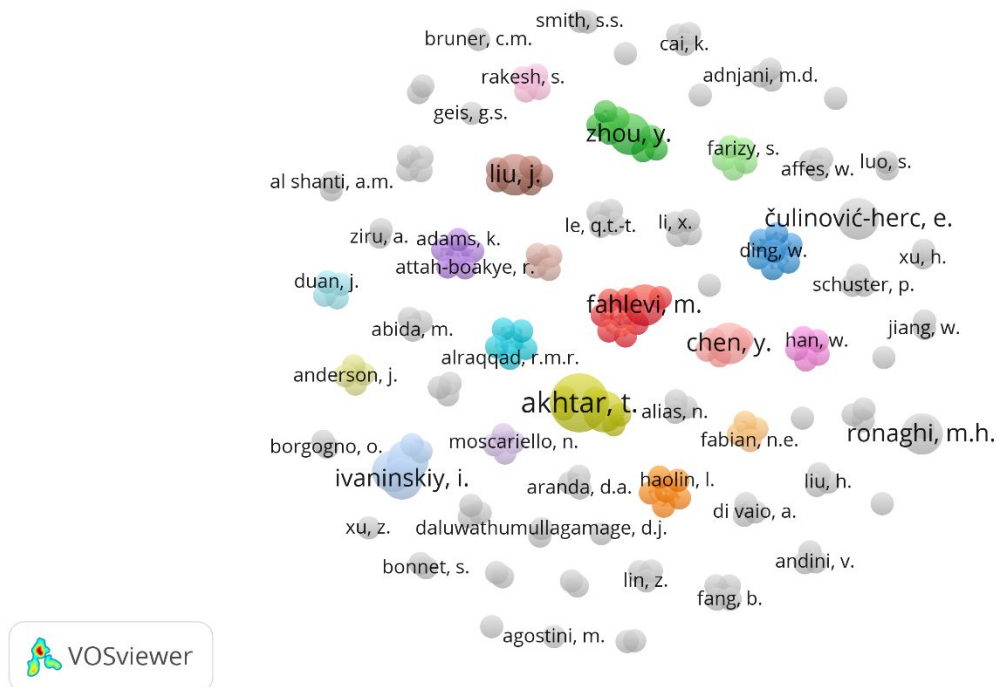


Figure 2: co- authorship of all authors

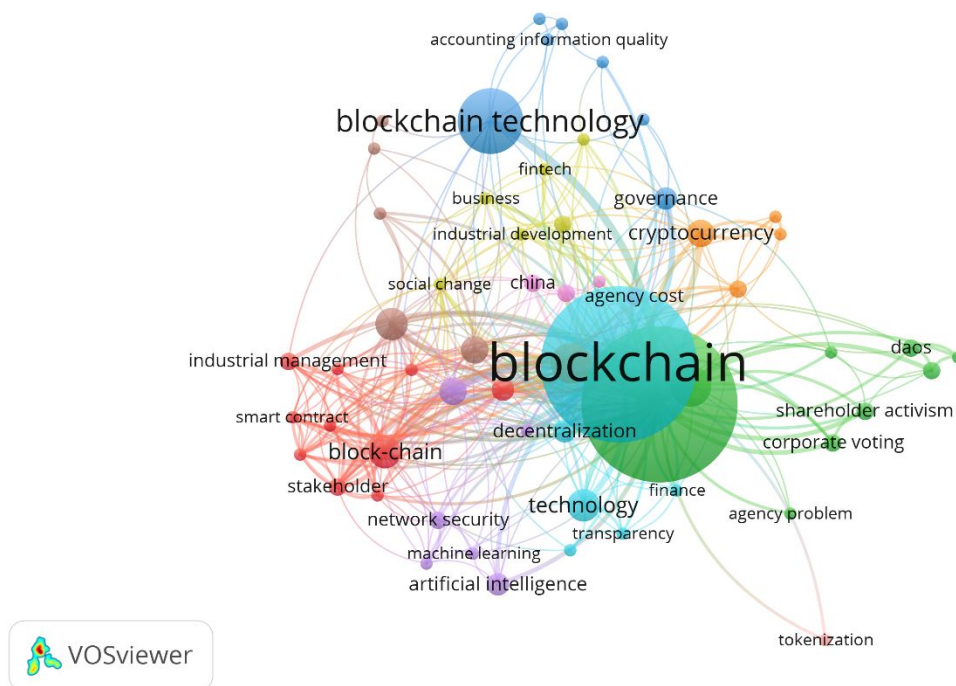


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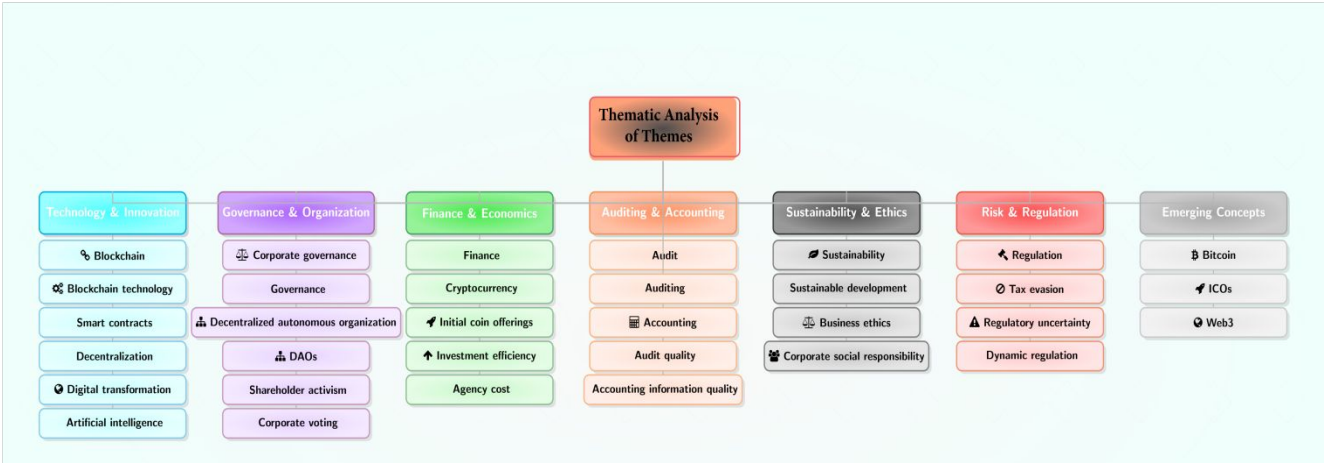


Figure 4 : Thematic visualization of emerging terms

Blockchain Technologies and Corporate Governance Reformation: An Agency Theory Perspective



1. Introduction

Emerging technologies are reshaping how corporations direct, monitor, and assure managerial conduct. These rapid technological advances are influencing how corporations enforce governance arrangements. In contrast to a manual, intermediary-based process, blockchain provides faster coordination and stronger verifiability, which are the capabilities long sought by corporate governance scholars to align dispersed investors, managers, and stakeholders (Fahlevi et al., 2022; Kimani et al., 2020a; Magnier & Barban, 2018). By providing transparent, tamper-evident records and programmable enforcement via smart contracts, blockchains promise to reduce information asymmetry, lower monitoring and enforcement costs and expand shareholder participation, which are the core layers in corporate governance theory and practice (Daluwathumullagamage & Sims, 2020a; Kaal, 2020; Yermack, 2015). These transformative capabilities of blockchain are reinforced by the dynamics of market demand that require continual realignment of skills, tasks, and technologies in corporate oversight and assurance (Akhtar et al., 2024a; Kimani et al., 2020a; Yusuf et al., 2023).

Across conceptual and empirical streams, scholars have linked blockchain-enabled mechanisms to improvement in disclosure and auditability, board oversight, shareholder voting, and ownership transparency. In these, agency theoretic arguments have been the most prominent and complemented by transaction cost economies, stakeholder, stewardship, and legal-institution perspective (Al Shanti & Elessa, 2023; Ivaninskiy & Ivashkovskaya, 2022a; Morrison et al., 2020; Mutamimah et al., 2023a; Yermack, 2015; Yusuf et al., 2023). Empirically, using regressions, structural equation modelling and case analysis, studies have empirically examined the impact of blockchain technology on International ESG firms using a principal-agent perspective (Ivaninskiy & Ivashkovskaya, 2022a), shareholders rights and participation (Jiménez-Gómez, 2023a; Xu, 2025), financial reporting quality, auditability and fraud reduction in banking section and other emerging markets (Al Shanti & Elessa, 2023; Ronaghi & Mosakhani, 2022). In contrast, Mutamimah et al. (2023) used a qualitative method to examine its impact on MSMEs in Indonesia.

Aside from the theoretical arguments and empirical studies, practically, market infrastructures are experimenting with distributed ledger technologies for share registries, proxy voting, and post-trade processing. Examples of corporate governance instances where corporations have employed blockchain technologies include the Delaware-authorized blockchain stock ledgers. Other

define abbreviations before use (is this the South Africa law mentioned in 2.2?)

Micro, Small and Medium Enterprises

reference

Estonia i-voting does not deploy blockchains; their e-government probably does (RIA)

reference

instances include Nasdaq’s proxy-voting pilots, such as Estonia, and exchanges initiatives, including the ASX CHESS replacement effort. These illustrate both the use of auditability and near real-time reconciliation and the practical and legal complexities of migrating from incumbent technological system (Akgiray, 2019; Fenwick et al., 2019; Lafarre & Elst, 2018). Decentralised autonomous organisation further stress-test corporate forms by shifting trust and decision rights toward code and token holders, raising open questions around accountability, control, and regulatory fit (Chambefort & Chaudey, 2024; Morrison et al., 2020; Sims, 2019; Tse, 2020). Hence this study synthesises prior scholarship to discuss how blockchain can transform corporate governance from the agency theory perspective.

Features

1.1 Blockchain Mechanisms Relevant to Corporate Governance

Literature has described how blockchain features can be included in corporate governance. Some of the features of blockchain include the immutability structure of being unable to easily delete transactions of its ledger systems. Transparent immutable ledgers for ownership and transactions (share books, insider-trade monitoring, real-time disclosure) are a blockchain features that targets the classic principal-agency frictions by improving verifiability and reducing opportunism (Caldarelli, 2020; Daluwathumullagamage & Sims, 2020a; Yermack, 2015; Yusuf et al., 2023). The transparency and immutability of blockchain can strengthen monitoring of insiders, clarify beneficial ownership and reduce reconciliation and settlement frictions (Agostini, 2024; Akgiray, 2019; Kimani et al., 2020a; Magnier & Barban, 2018).

Another blockchain feature relevant to corporate governance is smart contracts. Smart contract ~~automatic system~~ is used for vesting, dividends voting threshold and workflow compliance, potentially substituting for some ex-post auditing and ex-ante code-based enforcement (Caldarelli, 2020; Daluwathumullagamage & Sims, 2020a; Lafarre & Elst, 2018; Yermack, 2015). Moreover, the smart-contracts can potentially lower agency and transaction cost (Caldarelli, 2020; Yukhno, 2021).

say something like "are code transactions that get automatically executed online" or rephrase according to your writing style.

remove one of them

Also, blockchain is also relevant in on-chain, assisted voting and annual general meetings that lower participation frictions such as participatory costs, enhance audit trails, and enable faster decision making (Jiménez-Gómez, 2023b; Lafarre & Elst, 2018). This can increase turnout by interacting with cross-border identification and legal requirements (Jiménez-Gómez, 2023b; Lafarre & Elst, 2018; Magnier & Barban, 2018). Also, while blockchain can be used for tokenisation of equity and rights that can clarify beneficial ownership, improve registry accuracy and accelerate ownership transfers, it can also shift power and liquidity dynamics in ways corporate law and governance have yet to fully address (Agostini, 2024; Akgiray, 2019; Fenwick et al., 2019; Kimani et al., 2020a; Yermack, 2015).

The authors should discuss the impact of "vote delegation" here (they did not explain yet "the governance token"); voting powers can be transferred through token transfers: survey studies on it (useful whentechnical details areinvolved that some aren't familiar with, but could concentrate power and vote buying)

Foundational work argues that blockchain could "upend the balance of power" among relevant stakeholders such as the managers, investors, auditors and regulators by radicalising ownership transparency and automating enforcement (Kaal, 2020; Lafarre & Elst, 2018; Yermack, 2015). Empirical evidence reports a positive relationship between blockchain adoption and governance

outcomes (Al Shanti & Elessa, 2023; Ivaninskiy & Ivashkovskaya, 2022a; Mutamimah et al., 2023a; Ronaghi & Mosakhani, 2022). These studies show that board decision quality, shareholders' rights, perceived transparency, reduced agency conflicts and improvements in financial reporting quality across banking and listed firms contesting in multiple regions were all related to blockchain adoption and usage. For instance, (Ivaninskiy & Ivashkovskaya, 2022a) indicates there are reduced principal-agency conflicts in large US corporates. Other studies reveal there is improved transparency and auditability in banking, emerging markets and MSMEs settings. (Al-Shahamani et al., 2025; Mutamimah et al., 2023a; Ronaghi & Mosakhani, 2022)

Elimination of intermediaries is another feature of blockchain, which literature have shown to be relevant in corporate governance. This reduces the fees charged by the third parties, reduces transaction time and removes redundant activities, which translates to a faster and seamless corporate structure. While eliminating the need for intermediaries in several corporate governance-related activities, studies also argued that blockchain technologies have the potential of creating new links to other types of control points in transactions, which may be new forms of intermediaries. In permission versus non-permissioned architectures, oracles' dependencies and concentration risks can reintroduce intermediaries or create new control points, complicating the existing trust issues, thus, affecting governance efficacy (Caldarelli, 2020; Chambeft & Chaudey, 2024; Yin, 2023).

Furthermore, the decentralised automated organization in blockchain technologies exposes tensions between code-based control and legal accountability, suggesting hybrid techno-legal arrangements rather than wholesale replacement of traditional boards and fiduciary duties (Lafarre & Elst, 2018; Morrison et al., 2020; Sims, 2019; Tse, 2020). This is evidences in the DAO's relocation of decision rights into code and token-holders' votes, thus, exposing tensions around accountability and legal personhood (Chambeft & Chaudey, 2024; Morrison et al., 2020; Sims, 2019; Tse, 2020).

This study attempts to review the existing literature body on blockchain and corporate governance by synthesizing all the prior literature and discussing them in relation to agency theory. While studies have reviewed the intersect of blockchain and corporate governance, focusing specifically on discussion from the agency ^Perspective is a gap in literature. ~~For instance~~

For instance, Yermack (2015) is a conceptual mapping that serves as foundational paper which focuses on agency costs, ownership transparency, audit functions and shareholders' monitoring, which are all elements used in agency theories, but, the paper is old and it was published before the growth of literature on blockchain and corporate governance as shown in the section 4.1 of this paper which maps out the prior studies as found in scopus database. Also, Daluwathumullagamage and Sims (2020) wrote a systematic review of about 180 papers which focus on the application of mixed theories, including agency, stewardship, resource dependence and political theories, to map out a decentralised governance framework. The study is very broad, but this current paper has a

specific focus on blockchain and corporate governance by discussing it from only an agency theory perspective.

Since this a blockchain journal, I suggest things like cooperate governance, Agency Theory, stewardship,...etc should be explained first (problem definition), as not all readers are familiar with, then proceed to explain why blockchain features/characteristics fit here in addition to DAOs in general.

Agency theory is chosen as the discussion form for this study because of its ability to successfully interpret the corporate governance system and explain the dynamics existing between the relevant actors and stakeholders, such as the managers and shareholders. Also, based on the existing literature, the theoretical landscape in blockchain and corporate governance is uneven by its indication of the dominance of agency theory and scattered alternatives. For instance, many study uses agency theory, which is reflected in the constructs and hypotheses (Ivaninskiy & Ivashkovskaya, 2022a; Kaal, 2020; Ronaghi & Mosakhani, 2022; Yermack, 2015; Yukhno, 2021), while other theories, such as transaction-cost economies (Ronaghi & Mosakhani, 2022; Yukhno, 2021), stakeholder perspectives (Mutamimah et al., 2023a; Ronaghi & Mosakhani, 2022) and stewardship/resource-dependence views (Caldarelli, 2020; Daluwathumullagamage & Sims, 2020a), appear less consistently and often implicitly. Hence, given that this research is a literature review of past papers, this study uses the agency theory as the theoretical base for the discussion.

While the existing literature review has begun to map these strands and the governance mechanisms they target, consolidated study, theory-centered, mechanism-specific panorama remains incomplete (Daluwathumullagamage & Sims, 2020a). This study addresses that need through an integrated bibliometric and thematic review and thorough discussion section which clearly explains how agency theory can be used in describing the impact of blockchain technologies on corporate governance. Additionally, given that blockchain is an emerging technology, research on blockchain technologies and corporate governance is still in its early stages of development. Therefore, this study aims to provide an overview of the available studies on the SCOPUS database.

The Scopus database is also chosen for this study due to its indexing of journals that publish papers that match scientific rigour and standards. This complies with the rules expected of a scientific paper, justifying the choice of the Scopus database for this research. Also, its multidisciplinary breadth matches the topic's scope. Journals across business, finance, law, information systems, computer science and accounting, areas which are relevant to this research, are indexed and easily retrieved in the Scopus database. Its rich metadata, which are compatible with a bibliometrics analysis, make it suitable for this research. When compared with alternative databases such as Web of Science and Google Scholar, Scopus is preferred for this study. For instance, Google Scholar does not have consistent metadata suitable for bibliometric analysis and is subject to duplications, whereas Web of Science, although having rich metadata, has fewer indexed journals in areas examined by this research, with most of these journals also indexed in Scopus.

Data was analysed using the bibliometric and thematic review method using analysis applications such as Microsoft Excel, VOS viewer, and Harzing Publish or Perish. These choices of applications are chosen due to their ability to give a co-occurrence, co-citation and co-authorship

view of the data. Also, they aid the presentation of metrics related to the data analysis. Values such as the citation, affiliations of papers can be easily presented.

After the data analysis, this study discovered that the available articles started growing in an upward trend in 2022. This confirms that the area of research in blockchain inclusion in corporate governance is recent. The findings reveal the need for more empirical studies. Also, this study fills the gap in the literature by analysing the existing studies and discussing them in relation to the agency theory perspective. In the theory integration gap, agency theory dominates, alternative lenses such as transaction cost, stakeholders, stewardship and legal institutional theories are under-integrated into cumulated explanations and tests.

In addition, the review analysis shows there is an existing boundary and risk gaps in the mediating roles of permissioning, token/moderating concentration, and jurisdictional framework, and the new agency problems introduced by validators and DAOs (Caldarelli, 2020; Chambefort & Chaudey, 2024; Lafarre & Elst, 2018; Morrison et al., 2020; Tse, 2020). While these are conceptually noted, they are empirically thin in corporate concepts, which opens the suggestion for future studies.

Therefore, while reviews exist, a comprehensive theory-centered bibliometric and thematic synthesis that classifies the diversity of mechanisms, theories, methods, and geographies is still missing. Hence, this study conducts a science mapping and bibliometric analysis of blockchain-corporate governance literature, and also executes a structured thematic synthesis of past studies to address the specific research objectives outlined in the next subsection.

1.2 Research Objectives:

- 1) To conduct a performance and science-mapping analysis (Scopus) of blockchain-CG literature
- 2) To execute a structured thematic synthesis of the prior literature on blockchain and corporate governance.
- 3) To develop an integrated framework that maps blockchain functionalities to governance mechanisms and expected outcomes, explicitly aligning constructs with agency theory
- 4) To analyse how blockchain inclusion mitigates agency problems in corporate governance
- 5) To identify implementation conditions that maximise benefits while mitigating risks

This introduction section sets the stage for the overview of the blockchain's transformative impact on corporate governance. It highlights its roles in fostering corporate governance accountability, efficiency, and stakeholders' confidence in an increasingly digital world. The second section, literature review, explains the terms as defined in the papers in the dataset. The methodology section describes the systematic data collection. The objective of this study was achieved through examining all related articles. For this study, prior research on blockchain and corporate governance served as the data source. Data was collected using the SCOPUS database, based on

title, abstract, and keywords, to ensure full access to articles in this area. Seventy-five papers were included in the analysis, forming the dataset for this research. The findings section is presented in Section Four, providing an overview of the bibliometric analysis. This analysis reflects the publication trends, the top journals in which the dataset was published, the themes of publication, and the top productive articles, thereby achieving Objective One. Thematic analysis was conducted using VOSviewer and Microsoft Excel, where keywords were extracted and coded using LaTeX in the Overleaf application. This achieves the second objective of this research. The discussion section is presented in Section 5, which provides a detailed summary of the dataset. The discussion focused on the agency perspective in examining blockchain technology and corporate governance. This section achieves the objectives three and four of this research. Limitation of the study and suggestions for future studies was discussed in the final section.

2. Literature Review

2.1 Blockchain

Blockchain technology is a segment of the distributed ledger technology (DLT) that stores information and secures transactions. (Fahlevi & Pramesti, 2022; Yusuf et al., 2023). This enables users and relevant stakeholders to store information without time or country restrictions and ensures that, aside from those with access, no other person or entity can hijack the information. Hence, blockchain provides innovative, high-security solutions to organizations, ensuring the best way to secure all data, information, and transactions passed on the blockchain.

In corporate governance, blockchain is effective in promoting transparency, accountability, and efficiency within the management structure, the board members, and other relevant stakeholders (Akhtar et al., 2024b; Daluwathumullagamage & Sims, 2020b; Singh et al., 2020). Blockchain enhances efficient corporate governance through immutable record-keeping, real-time auditing, secure shareholder voting, and decentralized decision-making models. Through modeling, blockchain streamlines compliance for organizations and their diverse stakeholders by utilizing automated smart contracts, thereby enhancing transparency in reporting. This is exemplified by systems such as the Santander organization, which launched the first end-to-end blockchain bond, offering a faster, easier, and more efficient bond issuance process. While blockchain can revolutionize corporate governance, technology may not be enough. An instance is demonstrated by the Maersk blockchain-based digital logistics platform, Tradelens, which ceased operation in 2023 due to various factors, including commercial viability and insufficient global industry collaboration.

2.2 Corporate governance

Corporate governance refers to the globally significant role of facilitating and influencing organizational operations. It encompasses the comprehensive process of safeguarding shareholders' rights, given their primary ownership stake in a firm (Akhtar, 2024). Corporate

governance has been recognized as a framework that enhances economic stability and development, as evidenced by improved returns on assets, equity, and earnings per share. Furthermore, it contributes to organizational efficiency by reducing time wastage. Consequently, the integration of appropriate technologies into corporate governance has yielded a dual benefit: enhancing both the economic and financial landscape. As a holistic structure, corporate governance protects and manages the rights of shareholders and other stakeholders, including lenders, employees, customers, and the board of directors (Akhtar, 2024; Mutamimah et al., 2023a; Rien Agustin & Susilowati, 2019a).

Corporate governance serves as the foundational structure guiding an organization's systems, including the mechanisms through which firms are controlled and directed. Transparency and credibility form the core pillars of effective corporate governance, underpinning an organization's reputation, credibility, and operational efficacy (Daluwathumullagamage & Sims, 2020b). To uphold these foundational principles, regulations grounded in relevant legal frameworks provide the structure for business activities. For instance, legal regulations govern the interactions between the board of directors and shareholders, ensuring accountability and fairness.

Various countries have established their legal framework on which the corporate governance stands. For instance, in the African regions, Nigeria's Companies and Allied Matters Act (CAMA, 2020) emphasizes shareholder protection by enhancing transparency in corporate dealings. South Africa's Companies Act No. 71 of 2008 and the King IV Report on Corporate Governance for South Africa 2016 prioritize transparency and the integration of environmental, social, and governance (ESG) factors (Companies Act No. 71, 2008; King IV Report, 2016). In the Arab region, the United Arab Emirates' Federal Decree-Law No. 32 of 2021 on Commercial Companies mandates board diversity to foster inclusive governance (UAE Federal Decree-Law, 2021). Similarly, Saudi Arabia's Corporate Governance Regulations 2017, aligned with Vision 2030, promote accountability and ethical practices (Capital Market Authority, 2017). In Europe, the United Kingdom's Companies Act 2006 and the UK Corporate Governance Code 2024 enforce accountability and stakeholder engagement (UK Parliament, 2006; Financial Reporting Council, 2024). Spain adheres to the EU Transparency Directive (Directive 2004/109/EC), ensuring consistent reporting standards across member states (European Union, 2004), while Switzerland's Swiss Code of Obligations (CO) prioritizes long-term shareholder interests (Swiss Confederation, 1911, updated 2023). These legal frameworks can be enhanced by blockchain technologies, as exemplified by platforms like Maersk's TradeLens, which automate compliance and enable secure, transparent voting and reporting. However, challenges such as scalability and regulatory uncertainties persist (Akhtar, 2024; Mutamimah et al., 2023a; Rien Agustin & Susilowati, 2019a).

2.3 Agency Theory

Agency theory has been widely linked to corporate governance, focusing on minimizing the losses incurred by principals while agents manage the business on their behalf (Fahlevi & Pramesti, 2022). Compliance with the corporate governance structures has proven to be effective in strengthening the relationship between the principals and the agents, involving key entities such

as the board of directors, the audit committee, the auditor, and the ownership structure (Al Shanti & Elessa, 2023).

Effective corporate governance, when aligned with established structures, significantly increases the likelihood of achieving the ultimate organizational goal of maximizing shareholders' wealth. This perspective is reinforced by the Malaysian Code of Corporate Governance (MCCG, 2017). The integration of technological advancements, such as blockchain, into corporate governance further mitigates common agency problems, including earnings management, underinvestment, shirking, and excessively risky investments (Fahlevi & Pramesti, 2022; Ivaninskiy & Ivashkovskaya, 2022b).

should be moved with the rest of regulations in the previous subsection.

Therefore, this study posits that a relationship exists between blockchain technologies and corporate governance, utilizing agency theory as the theoretical foundation. For instance, blockchain's tamper-proof audit trail establishes a transparent ledger system, providing real-time access to financial information and relevant data. This transparency reduces agency issues such as information asymmetry, fostering greater trust and accountability between principals and agents.

3. Methodology

This section discusses data collection and analysis techniques. This study employs a bibliometric and thematic literature review as the methodology. The database used for this research is the SCOPUS database. This is due to the claim that SCOPUS holds the largest datasets of peer-reviewed journals. Hence, this study utilized the SCOPUS database for data collection. The search string used for the literature search is "corporate governance" AND blockchain. By using the title, abstract, and keywords, which aim to capture all related papers, the reason is not to limit papers to titles alone.

Based on the search using title, abstract, and keywords, 145 papers were identified as related. Then, the document type was limited to "articles". The focus is on capturing peer-reviewed papers published in journals. The articles were reduced to 75 documents. Without limiting the time era, this research captures all the available articles in corporate governance and blockchain. Although based on the analysis, the available studies are quite recent and can capture the expectations required for new technologies.

For generalization, all the 75 articles were included irrespective of languages, since all the abstracts were in or have been translated into English and can be cited based on the summary in the abstract. Hence, 75 articles were found to be relevant to the study. After each of the titles was examined one by one, none of the articles were deleted. Hence 75 articles were used for this literature review based on the date of article search, which is 9th of May, 2025. The summary for this methodology is presented in the flow diagram in Figure 1.

Data analysis was conducted using analysis tools such as VOS viewer, Harzing publish or perish, Microsoft Excel, and latex coding of themes, the result of the analysis formed the basis of the

discussion of this research. Based on the result of the analysis, this study discussed each of the variables, that is, technologies, blockchain, and corporate governance.

Figure 1: PRISMA Flow Diagram for Methodology

4. Result

Given the constant growth in global technologies, adopting technologies such as digital blockchain has become imperative to incorporate into the corporate governance landscape, maintaining the integrity and credibility of corporate governance and ensuring the stability and legitimacy of firms. An initial bibliometric analysis was conducted using Microsoft Excel to gather preliminary information. Afterwards, the thematic analysis was examined.

4.1 Bibliometric Analysis of the Dataset

The research question one on how blockchain-CG literature has evolved was answered in this section. The analysis of the bibliometric overview for the entire dataset was conducted. This was carried out to achieve the first objective of this study, “to conduct a performance and science-mapping analysis (Scopus) of blockchain-CG literature”.

4.1.1 Publication trend

First, the publication trend, that is, the number of publications per year, was analyzed. The result is presented in Table 1. Based on the results, the first papers emerged in 2018, indicating that this was when the literature started assessing what linkage can be found in the blockchain-corporate governance relationship. Seven papers were published prior to 2020. The number of publications started growing in 2020 but was significantly reduced in 2021. An explanation for this might be due to the COVID-19 lockdown and movement control measures, which resulted in many institutions globally being shut down. In 2022, there was a surge in publications, and the number continued to grow yearly after that. This can be justified as the era in which blockchain adoption is growing, and literature is starting to explore it.

Table 1: Documents by year of publication

Year	Documents
2018	3
2019	4
2020	9
2021	4
2022	12
2023	16
2024	20

2025	7
Total	75

However, based on the total of 75 literature available, when compared with other areas within management within the SCOPUS database, it can be deduced that blockchain in corporate governance is currently at its growth stage. Hence, this study suggests that more literature can explore it, conduct more empirical research on it, and examine it using various theoretical backgrounds and empirical-based research.

4.1.2 Analysis of Publication Language

An analysis on the publication language was also conducted. The result shows that the English language accounts for the communication medium through which 92% of the articles were published. This shows the wide acceptance of the English language globally. Also, articles published in the English language can be cited globally without a regional or language barrier. The other languages where the other 8% of the papers in the dataset are published include Croatian, Chinese, German, Italian, and Russian. This result is presented in Table 2.

Table 2: Documents by Languages

Language	Number of publications	Percentage (%)
English	69	92
Croatian	2	2.6
Chinese	1	1.3
German	1	1.3
Italian	1	1.3
Russian	1	1.3
Total	75	100

4.1.3 Analysis of highly cited papers

The citation of the dataset was analysed using Harzing's Publish or Perish application (Harzing POP). The result shows the top three most-cited articles (Kimani et al., 2020b; Murray et al., 2021; Yu et al., 2018a) with 145, 130, and 125 citations, respectively. All the articles having more than 15 citations are presented in Table 3. The high citation reflects the relevance of these papers to issues in blockchain integration in corporate governance. affected also by the publication year

Table 3: Highly cited papers (citations more than 15)

Cites	Authors	Title	Year
145	D. Kimani, K. Adams, R. Attah-Boakye, S. Ullah, J. Frecknall-Hughes, J. Kim	Blockchain, business and the fourth industrial revolution: Whence, whither, wherefore and how?	2020

130	A. Murray, S. Kuban, M. Josefy, J. Anderson	Contracting in the smart era: The implications of blockchain and decentralized autonomous organizations for contracting and corporate governance	2021
125	T. Yu, Z. Lin, Q. Tang	Blockchain: The Introduction and Its Application in Financial Accounting	2018
112	A. Di Vaio, R. Hassan, R. Palladino	Blockchain technology and gender equality: A systematic literature review <i>what this has to do with governance????</i>	2023
95	M. Fenwick, J.A. McCahery, E.P.M. Vermeulen	The End of 'Corporate' Governance: Hello 'Platform' Governance	2019
82	R. Lombardi, C. de Villiers, N. Moscariello, M. Pizzo	The disruption of blockchain in auditing – a systematic literature review and an agenda for future research	2022
81	E. Pimentel, E. Boulianne	Blockchain in Accounting Research and Practice: Current Trends and Future Opportunities*	2020
61	P. McGrath, L. McCarthy, D. Marshall, J. Rehme	Tools and Technologies of Transparency in Sustainable Global Supply Chains	2021
60	S.S. Smith, J.J. Castonguay	Blockchain and accounting governance: emerging issues and considerations for accounting and assurance professionals	2020
59	H. Singh, G. Jain, A. Munjal, S. Rakesh	Blockchain technology in corporate governance: disrupting chain reaction or not?	2020
56	M. Hanisch, C.M. Goldsby, N.E. Fabian, J. Oehmichen	Digital governance: A conceptual framework and research agenda	2023
52	M.H. Ronaghi, M. Mosakhani	The effects of blockchain technology adoption on business ethics and social sustainability: evidence from the Middle East	2022
41	T. Lambert, D. Liebau, P. Roosenboom	<i>there is also an 2025 Science Direct article entitled</i> Security token offerings"Unlocking the Potential of Security Token Offerings"	2022
32	D.J. Daluwathumullagamage, A. Sims	Blockchain-enabled corporate governance and regulation	2020
25	R.-T. Sun, A. Garimella, W. Han, H.-L. Chang, M.J. Shaw	Transformation of the Transaction Cost and the Agency Cost in an Organization and the Applicability of Blockchain—A Case Study of Peer-to-Peer Insurance	2020
24	W. Ding, X. Liang, J. Hou, J. Li, Y. Rouabah, Y. Yuan, F.-Y. Wang	A Novel Approach for Predictable Governance of Decentralized Autonomous Organizations Based on Parallel Intelligence	2023
24	M.H. Ronaghi	Contextualizing the impact of blockchain technology on the performance of new firms: The role of corporate governance as an intermediate outcome	2022
23	C. Qian, Y. Gao, L. Chen	Green Supply Chain Circular Economy Evaluation System Based on Industrial Internet of Things and Blockchain Technology under ESG Concept	2023
22	S. Wang, W. Wen, Y. Niu, X. Li	Digital transformation and corporate labor investment efficiency	2024
21	I. Ivaninskiy, I. Ivashkovskaya	Are blockchain-based digital transformation and ecosystem-based business models mutually reinforcing? The principal-agent conflict perspective	2022
20	J. Du, Y. Shi, W. Li, Y. Chen	Can blockchain technology be effectively integrated into the real economy? Evidence from corporate investment efficiency	2023
19	W.A. Kaal	Decentralized Corporate Governance via Blockchain Technology	2020
19	M.J. Rahman, A. Zirui	Clients' digitalization, audit firms' digital expertise, and audit quality: evidence from China	2023
18	M. Fahlevi, R.M. Pramesti	Blockchain technology in corporate governance and future potential solution for agency problems in Indonesia	2022
18	C.M. Bruner	Distributed ledgers, artificial intelligence and the purpose of the corporation	2020
18	N. Drljevic, D.A. Aranda, V. Stantchev	An Integrated Adoption Model to Manage Blockchain-Driven Business Innovation in a Sustainable Way	2022
17	F. Rien Agustin, D. Susilowati	Preventing corruption with blockchain technology (case study of Indonesian public procurement)	2019
17	A.M. Al Shanti, M.S. Eleassa	The impact of digital transformation towards blockchain technology application in banks to improve accounting information quality and corporate governance effectiveness	2023

4.1.4 Analysis of the most productive journals These numbers will probably change after updating the review

In addition, the analysis of the journals that published the most papers addressing blockchain integration in corporate governance was carried out. The result shows that the Journal of Knowledge Economy and Sustainability (Switzerland) published the most papers in our datasets, with three publications each. The other journals that published a minimum of two publications are presented in Table 4. This reflects that academics attempting to publish papers on blockchain integration in corporate governance can explore these journal sources for publication.

Table 4: Journal with the most publications in the dataset

Source	frequency
Journal of the Knowledge Economy	3
Sustainability (Switzerland)	3
Cambridge Law Journal	2
International Journal of Data and Network Science	2
Journal of the Knowledge Economy	2
Sustainability (Switzerland)	2
Technological Forecasting and Social Change	2

4.1.5 Analysis of the co-authorship

The co-authorship of all the authors was analysed using VOS viewer. This shows the overview of all the authors that were included in the dataset used for this research. The result is presented in Figure 2.

Figure 2: co- authorship of all authors

4.2 Thematic Analysis of Research

Thematic grouping was applied to group the articles. This analysis was carried out to achieve the second objective of this study, “to execute a structured thematic synthesis of the prior literature on blockchain and corporate governance”. Keywords were analyzed by coding using VOS viewer, Microsoft Excel, and LaTeX coding structure by using the Overleaf application. Each of the papers was examined. The VOS viewer analysis result shows that keywords such as decentralisation, smart contract, industrial development, shareholder activism, corporate voting, and others were explored by the papers in the dataset. The result from the VOS viewer is presented in Figure 3.

Figure 3: Co-occurrence of keywords using VOS viewer

Moreover, this study extends the thematic analysis by classifying the keywords from the dataset. The dataset in this context refers to the 75 papers used for this analysis. From the dataset, aside from the general themes like corporate governance, blockchain, and blockchain technologies, other themes that emerged are divided into sections, which are technology and innovation, governance and organization, finance and economics, auditing and accounting, sustainability and ethics, risks and regulation, and emerging concepts. In technology and innovation, themes like smart contracts, digital transformation, and artificial intelligence have been studied by the existing articles.

by the way, I did not find any paper in the Review about "DAOs" or DAOs governance

Under governance and organization, concepts such as decentralized autonomous organization (DAO), corporate voting, and shareholder activism have been examined by the articles in the dataset. Under finance and economics, concepts such as cryptocurrency, initial coin offerings, investment efficiency, and agency cost have been examined. Auditing, accounting, and accounting information have also been discussed. Some sustainability and ethics areas have been explored, such as business ethics, corporate social responsibility, and sustainable development. While most of these concepts have focused on the beneficial impacts of blockchain in corporate governance, areas relating to risks and regulations have also been explored. This includes concepts such as tax evasion, regulatory uncertainty, and dynamic regulation. Some concepts, such as ICOs and Web3, have also emerged in the literature. These themes were coded using latex coding technique in overleaf application and it is presented in Figure 4.

Figure 4 : Thematic visualization of emerging terms

5 Discussion

This study carried out an in-depth discussion on the dataset by classifying the discussion into sub-sections as presented in this section. This discussion aims to present a clear presentation of blockchain integration in corporate governance as it relates to agency theory. This helps in achieving the objective three, four and five of this study. The description of agency theory in explaining blockchain integration in corporate governance was discussed. This includes the potential of blockchain technologies in the mitigation of agency problems. Other discussions include the impact and associated risks of blockchain inclusion in corporate governance.

5.2.1 Blockchain Technologies and corporate governance

This discussion achieves the objective three of this study “to develop an integrated framework that maps blockchain functionalities to governance mechanisms and expected outcomes, explicitly aligning constructs with agency theory”. Studies have shown that blockchain technologies have the capacity of transforming corporate governance (Akhtar et al., 2024b; Daluwathumullagamage

& Sims, 2020b; Fahlevi & Pramesti, 2022; Singh et al., 2020). According to prior research, blockchain is segmented into two. The first is the public blockchain, the second is the permissioned blockchain. After this, the smart contract is discussed. Followed by the benefits of blockchain in corporate governance. This kind of terminology defining should not be placed in the discussion; should have been earlier in the paper while you should concentrate here on what the Review papers discussed. i.e., you should here categorize the papers you have found, the research directions in each aspects, the research gaps, the promising areas,...etc

1. Public blockchain

The public blockchain is the structure that provides a free market where there is free entry and free exit where there is no single owner or operator. There are several copies of ledgers and each user has access to one copy of the ledger (Akhtar, 2024). This is what hosts data such as Bitcoin and cryptocurrency. On this blockchain, there is freedom to view the blockchain. Most of these public blockchains are open-source, decentralized, and independent. For instance, in the case of cryptocurrency such as Bitcoin, miners verify the blockchain. Hence, the blockchain is like a ledger that records the total Bitcoin owned in a particular wallet. While names, bank accounts, and entities are used in banking blockchain, in Bitcoin, public keys are used to define it. Hence, anyone who has a public key can access and move Bitcoin. Also, if lost, public keys cannot be retrieved (Daluwathumullagamage & Sims, 2020).

2. Permissioned blockchain

This type of blockchain is also referred to as the private blockchain (Akhtar, 2024; Daluwathumullagamage & Sims, 2020a). Regarding the permissioned blockchain, there is a need for permission to view the blockchain, create transactions, and validate transactions. Under this structure, there is one or more operators and owners, and authorized users are provided with access to the ledger. Hence, the permissioned blockchain is run by consortia such as financial organisations and stock exchange platforms like NASDAQ. Hence, permissioned blockchain is not as decentralized as the public blockchain. They were not open-sourced either. For instance, some cryptocurrencies, such as Ethereum, also offer a permissioned version. It is not that Ethereum has a "permissioned" version, it has turned into Proof of Stake (POS) where people participate by staking a certain amount of ETH

Aside from cryptocurrency, these private blockchain technologies have also innovatively impacted organisations and the financial world (Fang et al., 2023; Schuster et al., 2020; Stolbov, 2018; Yusuf et al., 2023). This has served as an innovation that has the potential to disrupt how activities and transactions are being performed globally, into a more modern, updated, upgraded administrative structure. Hence, if incorporated into other facets of the social and economic structures, blockchain can potentially transform innovative ways of doing things globally. This also has the potential of making the world a truly global village by allowing the accessibility of transactions without time or place limitation. Being a chain, blockchain therefore has the potential of linking the world together, ~~as each transaction is structured like a block in the order of the chain, connecting one to another.~~ blocks are ordered in a chain, where each block contains a number of transactions. Also, such a statement should have appeared much earlier in the paper.

3. Smart Contract

In a permissioned blockchain, Smart contracts are a set of instructions ~~on a blockchain~~ you could say on a Transaction written in computer code, and they are core to harnessing the blockchain's capability (Cazazian, 2022). These smart contracts can be used to guarantee performance by counterparties involved in a

contract. Hence, smart contracts can be applied in corporate governance to ensure adherence to rules and regulations. This can also be enforced to resolve several legal issues that may arise. A smart contract can also be employed to analyze the performance of employees and the relevant compensation packages. Furthermore, smart contracts can mitigate agency costs in corporate governance and serve as a futuristic tool for ensuring ethical behaviors.

Also, smart contracts may be extended to be self-performing. For instance, stock options can be diversified into segments within the contract, which aims to mitigate risks and diversify investments over available options (Fahlevi & Pramesti, 2022). Hence, using smart contracts, blockchain technologies can provide a transparent and innovative structure for ownership management positions within the firm. This balances equity and debt structure. It also mitigates the risks of fraudulent practices committed within the firm. Hence, blockchain can be the agency theory's motivation for mitigating the agency risks that might arise within the management, such as underinvestment or shirking and over (high-risk).

A blockchain smart contract automates incentives and performance-based rewards, thus mitigating errors by enhancing verifiable track records (Cazazian, 2022). Thus, improving employee trust and confidence in the corporate governance systems. Although there are strong challenges currently regarding blockchain and corporate governance, such as adoption, many organizations are presently not in a position to adopt this disruptive technology yet. Hence, there is currently a challenge due to its scalability. Although there are benefits as reflected in the Maersk's Tradelens. Adoption of blockchain into governance frameworks such as the UK corporate governance code or South Africa's Companies Act may minimize agency costs and foster increased confidence and trust among stakeholders.

5.2.2 Agency Theory in Corporate Governance and Blockchain

Blockchain technology application in corporate governance has been examined through the lens of agency theory which seeks to address the misalignment between principals (owners) and agents (managers) (Ivaninskiy & Ivashkovskaya, 2022a). Given that the traditional corporate governance frameworks are increasingly becoming inadequate in the face of disruptive technologies and the global ongoing digital transformation, this necessitates the integration of innovative tools to enhance organizational resilience (Zhang and Luo). Blockchain facilitates this adaptation by enabling corporations to transcend jurisdictional boundaries, positioning them as global citizens with responsibilities extending to environmental stewardship, employment practices, and ethical conduct (Thompson, 2021).

The facilitation of organizational citizenship through technology extends the impact of agency theory beyond the conventional owner-manager relationship to encompass interactions between jurisdictional governments and corporations (Thompson, 2021). Beyond enhancing internal governance, blockchain's decentralized nature connects corporations to international networks, improving their adaptability to global systems (Rien Agustin & Susilowati, 2019b; Thompson,

2021). This improves their alignment with diverse regulatory environments across jurisdictions, fostering adaptability (Rien Agustin & Susilowati, 2019b; Thompson, 2021).

5.3 Mitigation of agency problems

This section addresses the research objective four and five which aims “to analyse how blockchain inclusion mitigates agency problems in corporate governance” and “to identify implementation conditions that maximise benefits while mitigating risks”. Based on the prior literature and papers, blockchain addresses agency problems, which are the conflicts between owners and management, management and government (citizen structure of organization), by improving transparency. This balances power sharing among stakeholders through enhanced voting structures, increased shareholders' participation, and reduced reliance on third-party intermediaries, thereby mitigating disparity in influence and information asymmetry (Ivaninskiy & Ivashkovskaya, 2022a; Rien Agustin & Susilowati, 2019b). Specific areas of improvement based on the literature include management-sponsored proposals, proxy contests, and shareholders-initiated proposals, where blockchain fosters equitable governance. Additionally, blockchain-based structures such as decentralized autonomous organizations (DAOs) offer novel regulatory solutions that enhance agency constructs by streamlining interactions and reducing agency costs (Kaal, 2020; Murray et al., 2021).

5.3.1 Economic and technological implications

The adoption of blockchain can lower agency costs by altering traditional contracting mechanisms within markets and organizations, though it may introduce new expenses related to implementation (Murray et al., 2021). Ivaninskiy and Ivashkovskaya (2022b) highlights that blockchain stimulates innovative business models, particularly in technology-driven ecosystems like finance, healthcare, and communications, to eliminate principal-agent conflicts. However, DAOs, as emerging economic models, may introduce vulnerabilities such as unpredictable code loopholes, which can be addressed through advanced technologies like computational experiments, parallel executions and artificial intelligence (Ding et al., 2023). These technological advancements further support the resolution of corporate governance challenges within DAO frameworks.

Furthermore, empirical evidence suggests that blockchain's most significant impact on agency-related issues occurs in sectors reliant on information technology, top among which are communication, finance, and health sectors, where it optimizes stakeholders' relationships and organizational efficiency (Ding et al., 2023). This dual role, improving internal agency dynamics while fostering global connectivity, positions blockchain as a transformation tool in modern corporate governance.

5.3.2 Technologies in Corporate Governance

Technologies have proven to be indispensable in corporate governance, especially in recent years. Organizational management is adopting innovative technological updates in its activities (Bello y Villarino & Bronitt, 2024; Cicu, 2024; Singh et al., 2020; Ustahaliloğlu, 2025). These technologies

are assistive in carrying out a seamless corporate governance structure. For instance, stock-related activities can now be managed effectively using AI.

In corporate governance, there is the board of directors, which consists of members who serve as the representatives of the shareholders. The minority shareholders are also duly represented. These members perform their responsibilities as fiduciarily as stated in the mechanisms guiding their responsibilities. They are also responsible for overseeing and implementing strategies for organisations. Hence, the board is like the meeting point between the shareholders and the management. In creating seamless interaction between these parties, it is imperative to mitigate agency problems that may arise.

Technologies such as artificial intelligence (AI) have been shown to assist in mitigating these agency risks (Akhtar, 2024; Awad et al., 2025; Fahlevi & Pramesti, 2022). The risks include factors such as underinvestment, where management shirks the firm's capital. Another problem is overinvestment, such as high-risk investments, which may put the firm at high risk, including insolvency and illiquidity. By including AI, such as blockchain and big data, in the firm's activities, such as financial analysis and predictive capability, ratios and percentages can easily be calculated using various parameters, thus fostering optimal investment decision-making.

In the board, there is an ethical need for the separation of power between the CEO and the chairman to improve the board's effectiveness. This separation of powers is aimed at reducing the concentration of too much power in a single person or position. The management is responsible for the daily operations of the organization, which aims to maximize the firm's wealth. The board, therefore, is tasked with the responsibility of ensuring that these activities align with the core goals and vision of the firm, which is to maximize the shareholders' wealth. AI assists with aligning the firm's activities with its vision, mission, and objectives. Activities that deviate from these can be detected automatically. For instance, the threshold level of transactions can be set, and once this threshold has been exceeded, AI can easily detect this.

Aside from this, the board ensures that the financial statements are prepared per the regulations stated by the legal frameworks. AI has aided in the preparation of financial statements. Activities such as large volume ledger entries can be automated using AI. AI can, therefore, transform the ways in which financial statements are prepared. After the financial statements are prepared, auditing and assurance of the financial statements are needed. The internal and external auditing are tasked with ensuring that the prepared financial statements are in line with the relevant regulations, detecting errors and mistakes, thus mitigating fraud. AI has also been proven as an assistive structure in the auditing and assurance services. The audit and assurance services also ensure that transparent information is provided to the stakeholders (Cazazian, 2022; Lombardi et al., 2022; Rahman & Ziru, 2023).

Through AI, auditing can now be automated, and real-time auditing of financial data can be implemented. This information presented to the relevant stakeholders must reveal the true and fair view of the firm's activities and transactions. Thus, AI integration in auditing can improve the independence of auditors and ensure the transparency of information. Transparency prevents agency risks, gives positive signals to the stakeholders, such as the customers, investors, and potential stakeholders (Cazazian, 2022; Rahman & Ziru, 2023), thus increasing the firm's wealth. All these are to ensure that the board is effective and independent, support the detection of managerial opportunism, and reduce the emergence of audit risks.

5.3.3 impacts of blockchain technologies in corporate governance

Blockchain, especially the permission blockchain, ensures secure data, information, and transactions where no entity can penetrate except those with the access key (Cazazian, 2022; Schuster et al., 2020). This has innovatively reduced risks and fraud experienced by organizations. Also, the financial data has been able to be kept secure. This has helped the going concern of organizations prevent insolvency and bankruptcy of several firms. Aside from this, the fraud signal that might be sent to the relevant stakeholders is minimized. Therefore, serving as a good signal to relevant stakeholders, including investors and potential stockholders. Entities and individuals invest in systems perceived to be transparent and secure. One of the explanations of agency theory is to ensure the wealth maximization of shareholders, by ensuring the smooth running of the firms by the management (Fahlevi & Pramesti, 2022; Ivaninskiy & Ivashkovskaya, 2022b). Hence, blockchain technology aims to fulfill this goal by ensuring that the organization's transactions are transparent and secure. This will assist in reducing the risk of fraud and ensure that the true and fair view of the financial and non-financial data is presented to the stakeholders.

Also, blockchain technologies have transformed how users of financial data retrieve the data by making it easier to get up-to-date and the latest needed information (Cazazian, 2022; Schuster et al., 2020; Yu et al., 2018b). Thus, fostering trust in the firm and its data. For instance, stakeholders can access the latest accounting data and make their own analyses and ratios, which can be used for investment and economic purposes, forecasting, and academic purposes. Hence, these relevant stakeholders can make informed decisions based on their own accounting judgements such as accounting for depreciation, interest rate, inflation rates, inventory revaluations and other financial readjustments (Fang et al., 2023; Pimentel & Boulianne, 2020; Stolbov, 2018; Yusuf et al., 2023).

Aside from this, blockchain technologies can transform an organization into a futuristic company by bringing novelty into the firm's systems and structures, thus transforming organizations into a fully digitalized network of stakeholders and activities. Nowadays, blockchain technologies are highly effective in empowering the design of innovative business structures, models, products, and

services, among others (Ivaninskiy & Ivashkovskaya, 2022b; Marie Younis, 2024; Qian et al., 2023). Organizations are adopting these technological structures due to their strength in building a strong business model. Blockchain is also a foundation on which organizational innovation can be built. Also, the value chain innovation provides a strong security structure for organizational data management.

Only the last paragraph of this section summarizes the risks, the rest lacks enough useful info

5.3.4 Risks of Blockchain in corporate governance

While blockchain has been instrumental to the success of business models, there is no guarantee of their success (Agostini, 2024; Mutamimah et al., 2023b). This may be due to the constant advancements in technologies, which make the existing blockchain technology obsolete, sometimes, causing high failure and low success of business models. Hence, blockchain has to be updated by organizations to meet the ever-growing demand in the business structure. Up to 2019, studies have shown a failure rate as high as 90% in blockchain technologies. Some of the reasons given were that there is no existing business model available for managing the adoption of blockchain technologies in the most sustainable ways. Because, it is a form of trial and error, leading to high failure rate. Thus, technological failures in blockchain have led to several economic downturns. These negative effects can be found in poor risk management, poor financial management, human relationship chaos, and poor performance in the corporate structure. Also, compared to the present, there is a considerable gap in blockchain technologies over the past few years and the present. Due to these gaps, it is estimated that almost all the past blockchain structures need to be overhauled and replaced with the updated technological structure of blockchain to increase capability, scalability, speed, effectiveness, and performance.

Blockchain technologies have been described as a disruptive technology capable of altering the socio-economic situations and activities, such as the financial industry activities and capability (Daluwathumullagamage & Sims, 2020a). As described as disruptive technologies, blockchain has the capability of challenging traditional and technological activities globally. In corporate governance, blockchain has been discussed in its capability of altering corporate governance performance. Studies have shown that blockchain technology has the capacity to lower corporate governance costs, improve reporting structure, increase liquidity, and provide greater transparency in the governance structure. For instance, studies on the banking sector reveal that the combination of various ledgers in blockchain technology can speed up processes, reduce costs, and provide a smooth operation process. So also, in the accounting processes, blockchain technologies result in triple-entry accounting, real-time auditing and assurance, reducing earnings management and other agency-related challenges (Daluwathumullagamage & Sims, 2020a; Drljevic et al., 2022).

In blockchain technologies, there is a need to identify core areas of risk. According to KPMG (2018), these risk areas include access user management, authorization and provisioning management, data management, interoperability, scalability and performance, change management, privacy, data and transaction security. Others include business efficiency, maintenance, storage, network load, reliability, architecture, standardization, computational complexity (Drljevic et al., 2022).

6 Conclusion, limitations, and suggestions for future studies

While blockchain has been revealed to be important in the technological transformation of corporate governance, especially based on the agency perspective, this study examined five research objectives that analysed the existing research using bibliometric and thematic analysis techniques. The result of this research discusses blockchain technology's inclusion in corporate governance. The result of this research has proven that blockchain technologies are important for mitigating existing agency problems, although new issues may arise in the future, such as coding and technology-related problems that may arise in blockchain-influenced corporate governance. Also, problems may arise based on regulations and from a legal perspective. Moreover, while blockchain technology may reduce agency-related problems and costs, new costs such as the cost of investment and maintenance of the blockchain technology may arise in the future. There is a need to mitigate these future problems that may arise in blockchain inclusion in corporate governance. Hence, future studies may examine how to mitigate blockchain-influenced agency problems, the risks, and the associated security challenges.

Aside from this, while this study focused on the analysis of prior articles using one database (SCOPUS), future studies are encouraged to conduct various empirical studies using primary and secondary data, which extends beyond the academic-related databases. Since this study is restricted to one academic database, future studies that intend to analyse past studies may extend their analysis to multiple databases, such as Web of Science and other academic sources, to gather all the relevant studies on corporate governance and blockchain. Finally, future studies may examine the country-specific impact of blockchain on corporate governance, and comparisons can be made using various sectors and countries that have adopted blockchain technologies in their corporate governance systems.

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